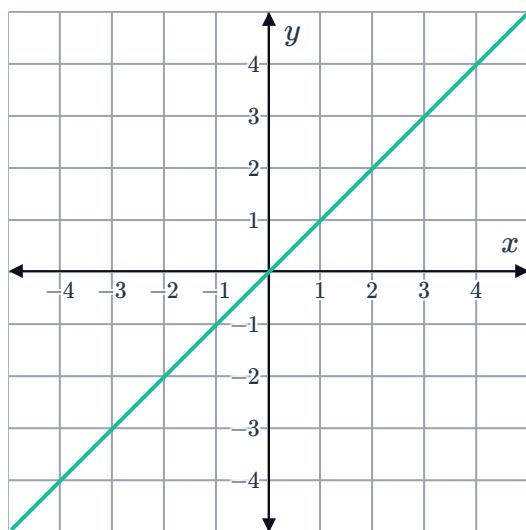


Equations of lines

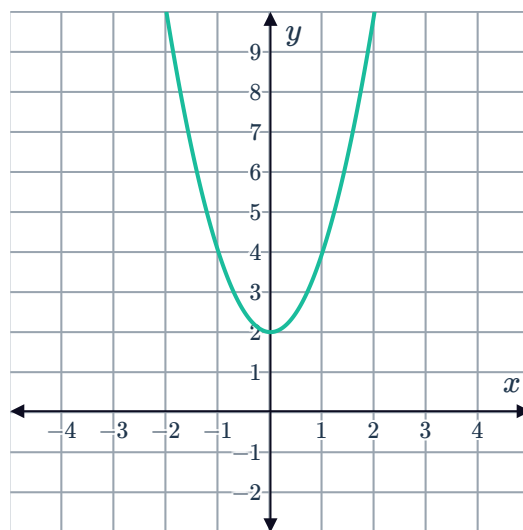


1 Determine whether the following graphs are linear:

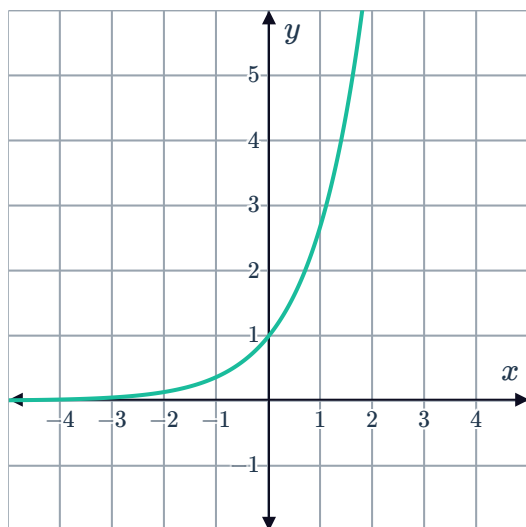
a



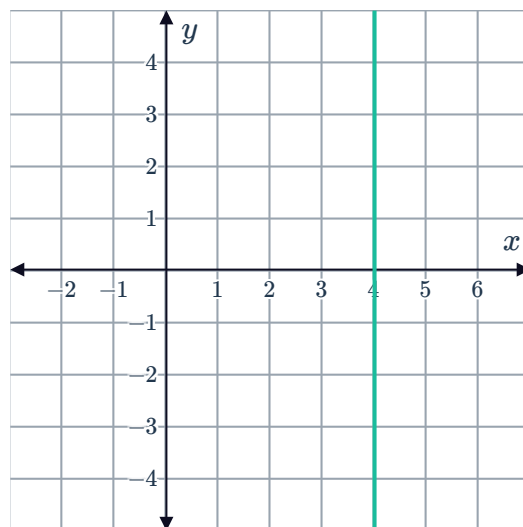
b



c



d



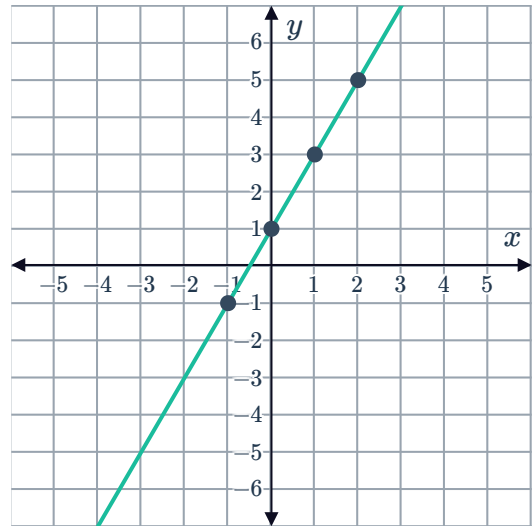
2 Consider the following graph:



- a Complete the table of values for the points shown on the line.

x	-1	0	1	2
y				

- b State the values of the gradient m , and the y -intercept, c .
- c Given that the linear relationship is in the form $y = mx + c$, write the equation of the graph.
- d Hence find the coordinates for the point on the line when $x = 21$.

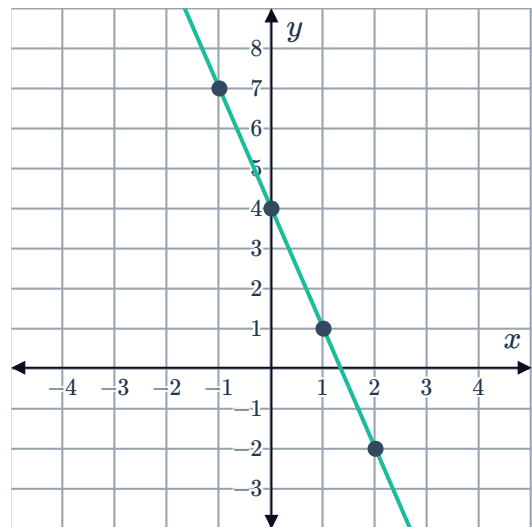


3 Consider the following graph:

- a Complete the table of values for the points shown on the line.

x	-1	0	1	2
y				

- b State the values of the gradient m , and the y -intercept, c .
- c Hence write the linear equation for the graph.
- d Find the coordinates for the point on the line where $x = 27$.



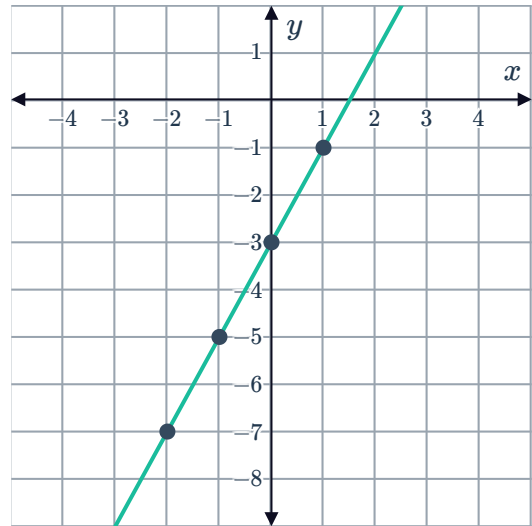
4 Consider the following graph:



- a Complete the table of values for the points shown on the line:

x	-2	-1	0	1
y				

- b State the values of the gradient m , and the y -intercept, c .
- c Hence write the equation for the graph.
- d Find the value of y when $x = 40$.

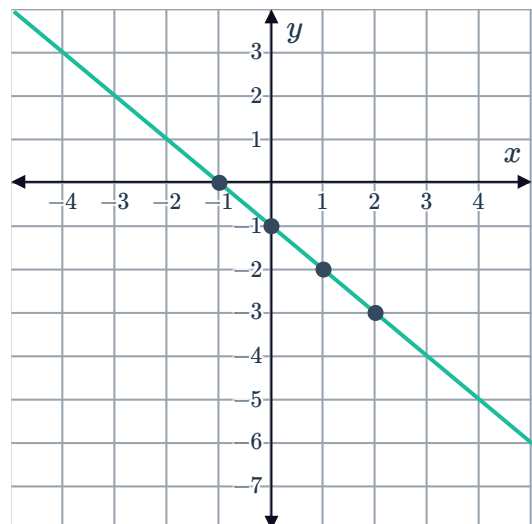


5 Consider the following graph:

- a Complete the table of values for the points shown on the line:

x	-1	0	1	2
y				

- b State the values of the gradient m , and the y -intercept, c .
- c Hence write the equation for the graph.
- d Find the value of y when $x = 19$.



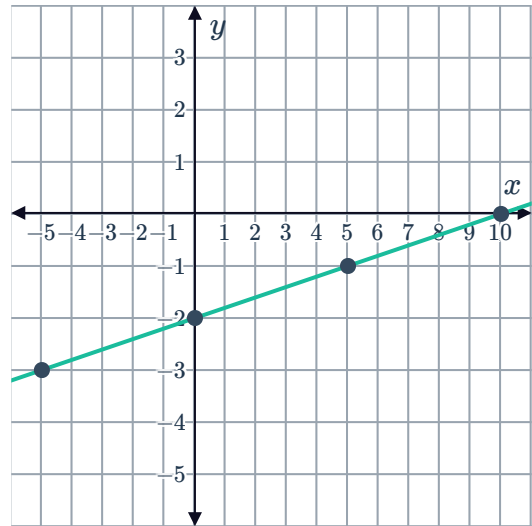
6 Consider the following graph:



- a Complete the table of values for the points shown on the line:

x	-5	0	5	10
y				

- b When x increases by 5, by how much does y increase?
- c Hence when x increases by 1, by how much does y increase?
- d State the values of the gradient m , and the y -intercept, c .
- e Hence write the equation for the graph.
- f Find the coordinates for the point on the line where $x = 35$.

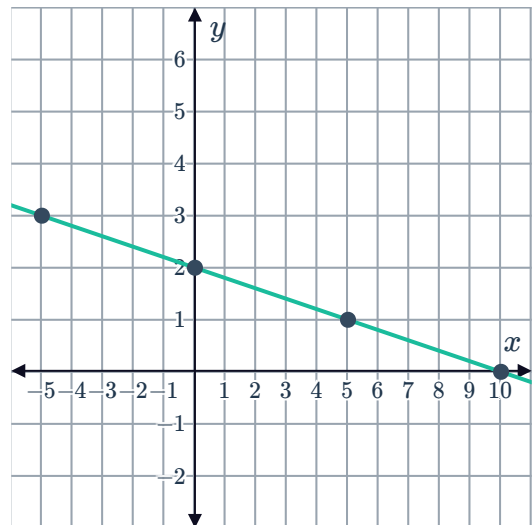


7 Consider the following graph:

- a Complete the table of values for the points shown on the line:

x	-5	0	5	10
y				

- b State the values of the gradient m , and the y -intercept, c .
- c Hence write the equation for the graph.
- d Find the coordinates for the point on the line where $x = -45$.



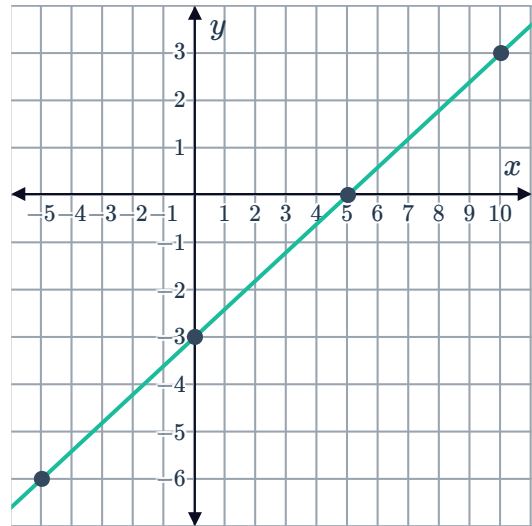
8 Consider the following graph:



- a Complete the table of values for the points shown on the line:

x	-5	0	5	10
y				

- b State the values of the gradient m , and the y -intercept, c .
- c Hence write the equation for the graph.
- d Find the coordinates for the point on the line where $x = 45$.

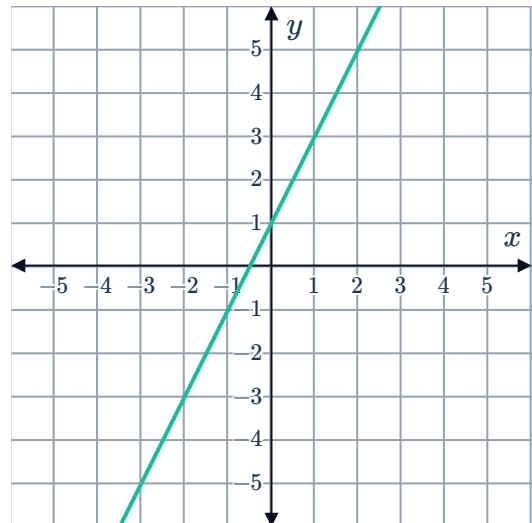


9 Consider the following graph:

- a Complete the table of values for the line shown in the graph:

x	0	1
y		

- b As x increases by 1, by how much does y increase?
- c State the values of the gradient m , and the y -intercept, c .
- d Hence write the equation of the line.
- e Find the y -value of the point on the line where $x = 22$.



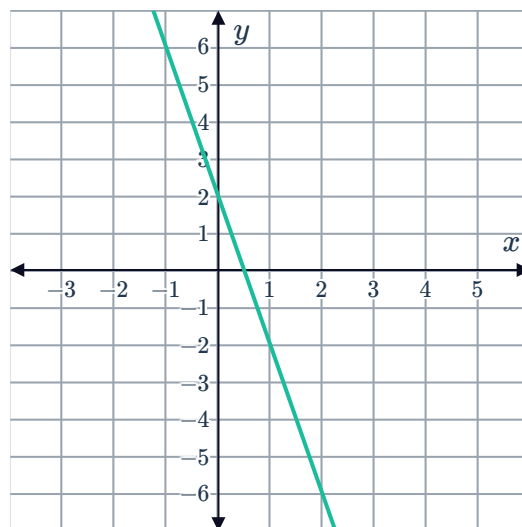
10 Consider the following graph:



- a Complete the table of values for the line shown in the graph:

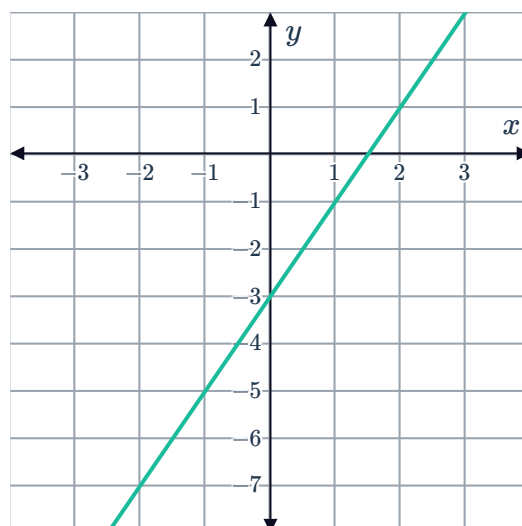
x	0	1
y		

- b As x increases by 1, by how much does y decrease?
- c State the values of the gradient m , and the y -intercept, c .
- d Hence write the equation of the line.
- e Find the y -value of the point on the line where $x = 27$.



11 Consider the following graph:

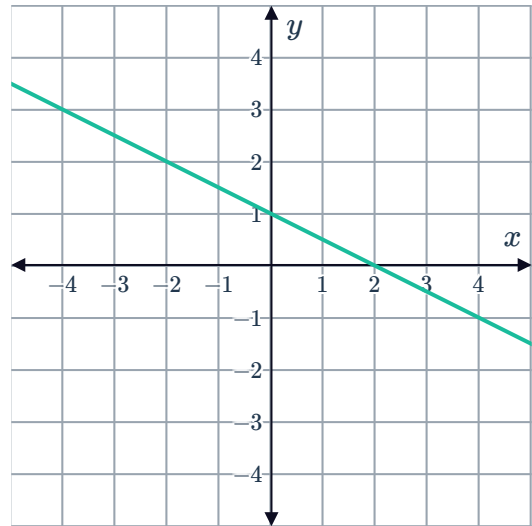
- a As x increases by 1, by how much does y increase?
- b State the values of the gradient m , and the y -intercept, c .
- c Hence write the equation of the line.
- d Find the coordinates for the point on the line where $x = 50$.



12 Consider the following graph:



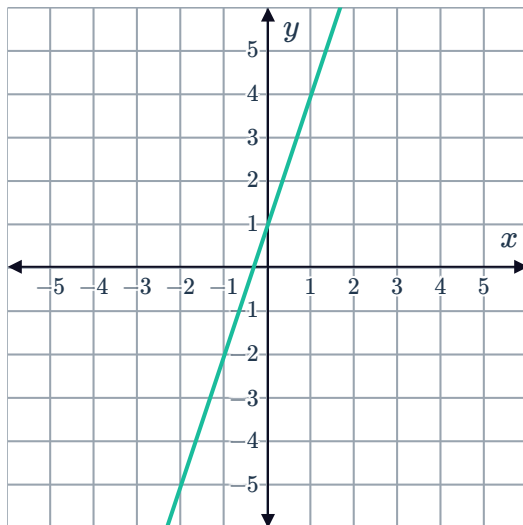
- a As x increases by 2, by how much does y decrease?
- b Hence as x increases by 1, by how much does y decrease?
- c State the values of the gradient m , and the y -intercept, c .
- d Hence write the equation of the line.
- e Find the coordinates for the point on the line where $x = -14$.



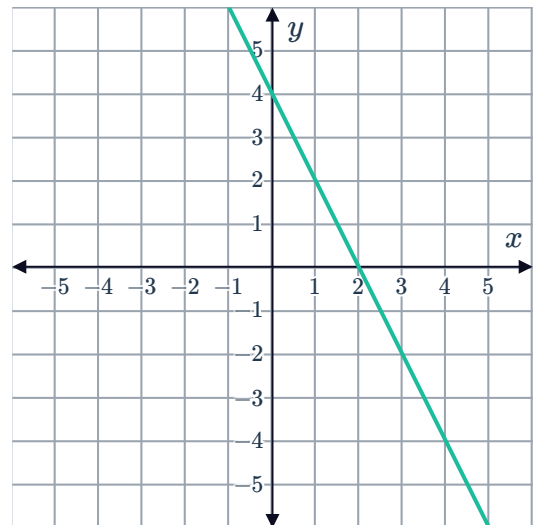
13 For each of the following graphs:

- i State the coordinates of the y -intercept.
- ii Find the gradient of the line.
- iii Write the linear equation expressing the relationship between x and y .

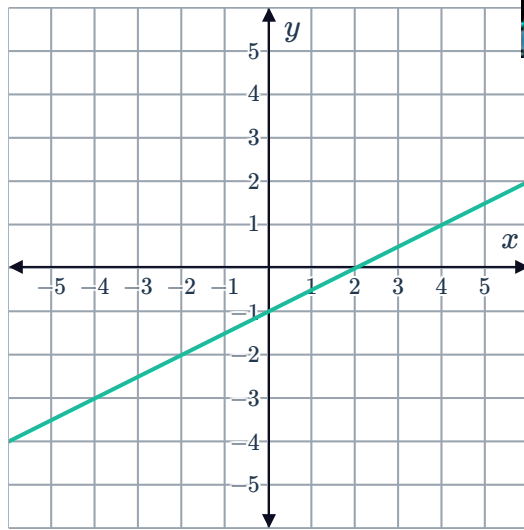
a



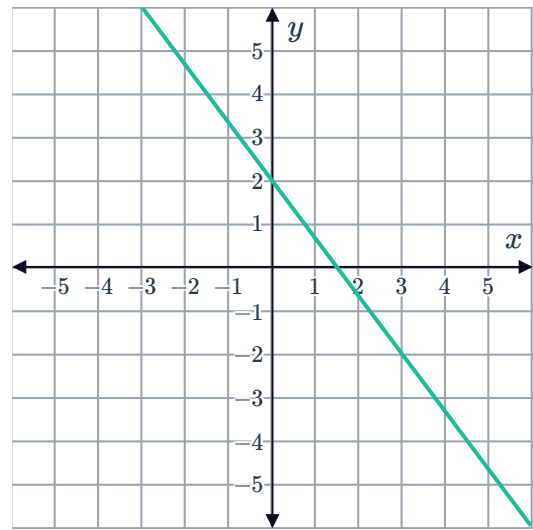
b



c

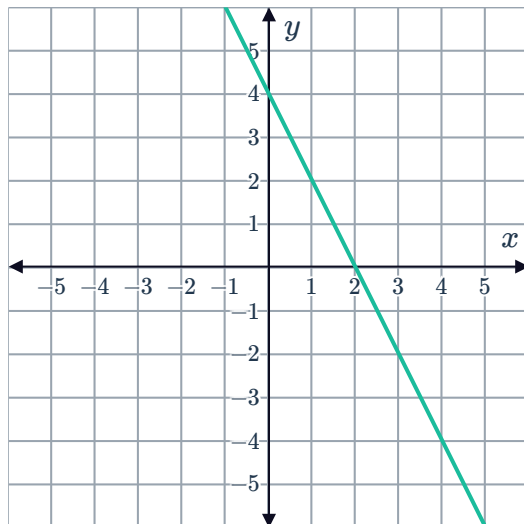


d

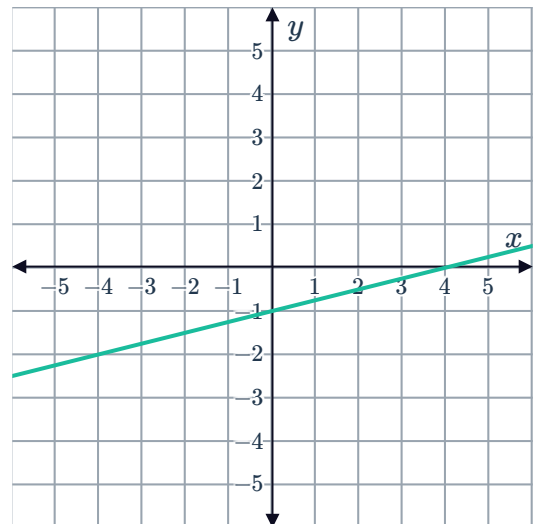


14 Find the equation expressing the relationship between x and y , for each of the following graphs:

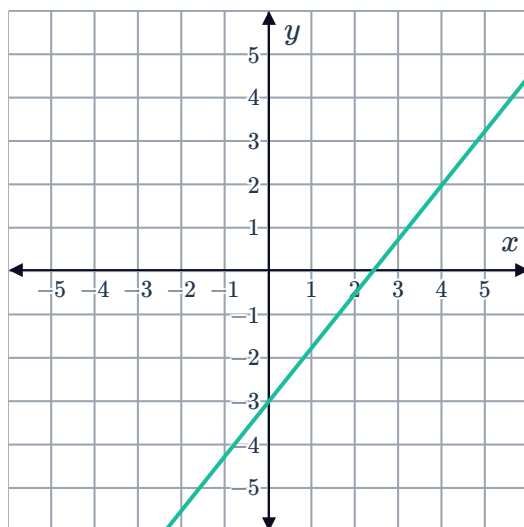
a



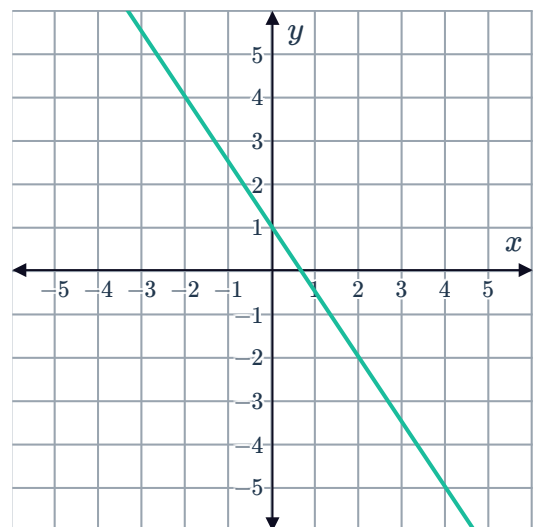
b



c



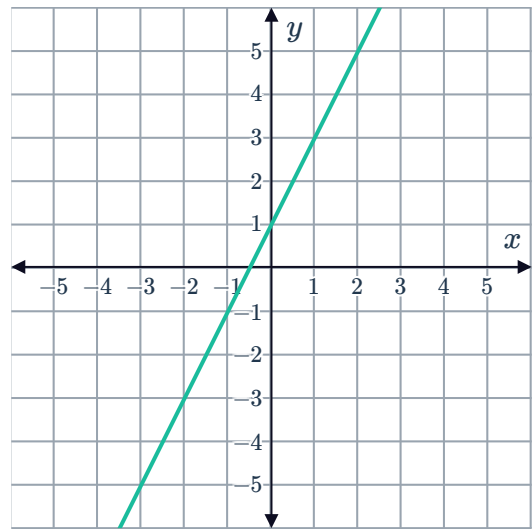
d



15 Consider the following graph:

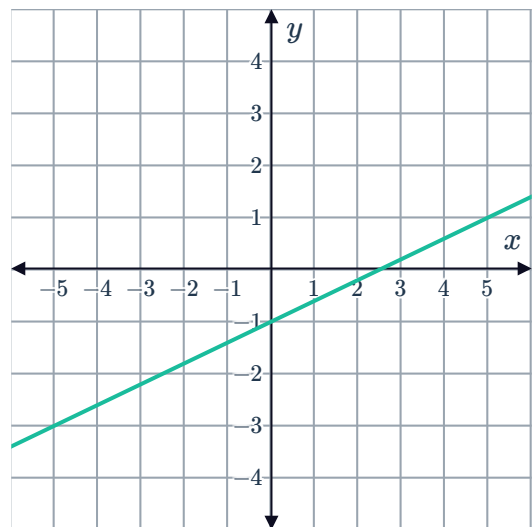


- a Find the linear equation expressing the relationship between x and y .
- b Hence find the coordinates for the point on the line where $x = 22$.



16 Consider the following graph:

- a Find the linear equation expressing the relationship between x and y .
- b Find the coordinates for the point on the line where $x = 25$.



17 Find the equation of the line, given the following information:

- a The line has y -intercept of 4, and gradient of -3 .
- b The line has y -intercept of -2 , and gradient of 9.
- c The line contains the point $(0, 7)$ and has gradient of -1 .
- d The line contains the origin and has gradient of $\frac{3}{4}$.
- e The line contains the origin and the point $(1, 3)$.
- f The line contains the origin and the point $(1, -5)$.
- g The line contains the points $(0, -5)$ and $(1, 1)$.
- h The line contains the points $(0, 2)$ and $(2, 6)$.